Lie Groups — Journals

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We will discuss several definitions of a smooth manifold on the special case that the manifold X is a euclidean manifold (embedded manifold), namely $X \subseteq \mathbb{R}^n$ for some $n \in \mathbb{N}$. Note that $\mathbb{R}^n$ is a T4 topological space of 2nd countability, therefore if A is an atlas on X then (X, A), or for the sake of convenience A alone, is a smooth manifold. When discussing embedded manifolds, there are several definitions, they are generally give substance to the concept of a manifold.

The first definition is Guillemin & Pollacks (GP) definition, as shown in their book "Differential Topology".

Definition 1.1 (GP smooth manifold)**.** Let $M \subseteq \mathbb{R}^n$ be a set and $k \in \mathbb{N}$. We say that M is a k-dimensional smooth manifold if that for any point $p \in M$ there is an open neighborhood $p \in U \subseteq \mathbb{R}^n$ and an open set $W \subseteq \mathbb{R}^k$ with function $F : U \to W$ diffeomorphism, meaning that its smooth and invertible. We define the function $f = F|_{M \cap U}$ as a smooth function from open subset of the manifold M.

Another version which we consider is defined by its parameterizations,

Definition 1.2 (Locally smooth parameterization manifold)**.** a set $M \subseteq \mathbb{R}^n$ together with a $k \in \mathbb{N}$ is called a smooth manifold if for any $p \in M$ there is an open set $U \subseteq M$ (open in the sub-topology of $\mathbb{R}^n$ induced by M), and an open set $W \subseteq \mathbb{R}^k$ such that there exists a function $\varphi : W \to U$ smooth (in the standard sense), open, invertible and with rank $D_x = k$ for any $x \in W$, namely a smooth local parameterization.

While both definitions may seem different, it can be shown using the inverse function theorem that they are equivalent.

Whitney embedding theorem states that any smooth manifold (X, A) can be embedded in $\mathbb{R}^{2k}$ space where $k = \dim(A, X)$, giving us motivation to compare our definition of a manifold in euclidean space to the definitions stated above.

Proposition 1.3. *If $M \subseteq \mathbb{R}^n$ is a GP k-dimensional smooth manifold and for any $p \in M$ let $f_p : U_p \to W_p$ be a diffeomorphism, where $U_p \subseteq M, W_p \subseteq \mathbb{R}^k$ open. Then $A = \{f_p \mid p \in M\}$ is an atlas, and in particular (M, A) is a smooth manifold.*

Proof. Clear by our definition of f_p as a restriction of a smooth function F_p and compositions. □

This result together with Whitney embedding theorem lead us to the conclusion that a manifold is a smooth manifold iff its a GP manifold (up to embedding).